

SubjectCode/Subject Name :CS4304ARTIFICIAL INTELLIGENCE

Year&Semester:IIYEAR/IIISEM

Slot.	Unit	Questions	Grade	Mark	Sign
I	UNIT V REINFORCEMENT LEARNING AND AI APPLICATIONS	1. Explain the SARSA algorithm with its update rule. Why is it considered an on-policy algorithm?	A		
		2. Explain the Q-Learning algorithm with its update rule. Why is it considered an off-policy algorithm?	A		
		3. Compare the assumptions, advantages, and disadvantages of Dynamic Programming and Temporal Difference learning methods.	A		
		4. Explain the Markov Decision Process (MDP) framework in full detail. Define value functions and derive the Bellman Optimality Equations. Explain how these equations form the foundation for optimal control in RL.	A		
		5. Describe the Policy Iteration algorithm in complete detail. Include the iterative policy evaluation and policy improvement steps. Prove that policy improvement monotonically leads to a better policy and eventually to the optimal policy	A		
		6. Discuss the ethical implications of AI in two of the following areas: Facial Recognition, Autonomous Weapons, Algorithmic Bias in Hiring. Propose a framework for the responsible development and deployment of AI in these areas	B		
II	UNIT IV PLANNING AND QUANTIFYING UNCERTAINTY	1. In a Bayesian Belief Network (BBN), consider the following weather scenario: - If the sky is Cloudy, there is a higher chance of Rain. - If it rains, the Sprinkler is usually turned off, otherwise it may be on. - The Grass becomes wet if either the Sprinkler is on or it is raining. Using the example "Cloudy and Rainy," answer the following: 1. Draw the Bayesian Belief Network with nodes: Cloudy, Rain, Sprinkler, WetGrass. 2. Write the conditional probability tables (CPTs) for the nodes Rain and WetGrass. Compute the probability that the grass is wet given that it is cloudy.	A		
		2. Define Uncertainty in AI. What are its causes? Explain how probabilistic reasoning addresses uncertainty.	A		
		3. Discuss Hierarchical Planning in Autonomous Driving with suitable examples.	A		
		4. Explain the applications of Multi-Agent Planning in Robotics, Traffic Management, Supply Chain, and Smart Grids.	A		
		5. Discuss Classical Planning. Explain Forward State Space Planning (FSSP) and Backward State Space Planning (BSSP).	A		
		6. Spam Email Detection using Bayes' Theorem An email filtering system uses the word "win" to classify emails as spam or not. The following probabilities are given: $P(S) = 0.2 \rightarrow$ Prior probability that any email is spam $P(H) = 0.8 \rightarrow$ Prior probability that any email is not spam (ham) $P(W S) = 0.6 \rightarrow$ Probability that the word "win" appears in a spam email $P(W H) = 0.1 \rightarrow$ Probability that the word "win" appears in a non-spam email Using Bayes' Theorem: Calculate the probability that an email is spam given that it contains the word "win." Show all calculation steps clearly. Write the final conclusion for the classifier.	B		

III	UNIT I INTRODUCTION TO AI	1.	Explain Breadth-First Search and Depth-First Search with examples.	A		
		2.	What are admissible and consistent heuristics? Why are they important in A* search?	A		
		3.	Formulate a delivery routing problem as a state-space search.	A		
		4.	Evaluate the efficiency of A* vs. Greedy Best-First Search. Which performs better and why?	A		
		5.	Design a PEAS description and a corresponding agent architecture for an AI-based traffic control system in a smart city. Justify your design.	A		
		6.	Explain the types of environment.	B		
IV	UNIT II ADVANCED SEARCH STRATEGIES	1.	Explain the complete working of a Genetic Algorithm with an example, covering initialization, selection, crossover, mutation, and termination.	A		
		2.	Compare and contrast Particle Swarm Optimization and Ant Colony Optimization in terms of inspiration, algorithm design, and applications.	A		
		3.	Describe alpha-beta pruning in detail. Show its working using a game tree and explain how pruning reduces computational cost.	A		
		4.	Explain in detail the various local search algorithms, including hill climbing, simulated annealing, tabu search, and genetic algorithms. Compare their performance in solving optimization problems.	A		
		5.	With suitable examples and diagrams, explain the various methods of solving CSPs, including backtracking, forward checking, arc consistency, and local search using the min-conflicts heuristic	A		
		6.	Explain in detail about PSO	A		
V	UNIT III KNOWLEDGE REPRESENTATION	1.	Discuss forward chaining in First Order Logic with an example knowledge base.	A		
		2.	Discuss backward chaining in First Order Logic with an example query and rules.	A		
		3.	Represent the following in FOL and perform unification: “All men are mortal. Socrates is a man. Therefore, Socrates is mortal.”	A		
		4.	Explain the architecture of a Knowledge- Based System with a neat diagram.	A		
		5.	Explain different laws of inference in propositional logic with truth table examples.	A		
		6.	Explain the laws of inference	B		